

Deep research on the search instructions and embedded research tasks in the provided files

Executive summary

Two user-provided documents define the research programme. The first (a comprehensive project report) frames an ongoing research agenda on neural successive-cancellation decoding for two-user MAC polar codes, explicitly calling out two “future directions” as flagship workstreams: closing the performance gap at larger block lengths on continuous-output channels (especially at and beyond $N=256$) and, more ambitiously, parallelising training to reduce sequential depth and unlock scaling to longer codes. [filecite\turn0file0](#)

The second (a “research brief” / prompt) is a structured search-and-ideation assignment: it specifies what must be explained, what kinds of literature must be looked for (group convolution, tropical algebra, tensor decompositions, parallel algorithms, etc.), what must be delivered (ranked approaches + feasibility analysis), and what constraints any solution must satisfy (notably: must handle Class B interleaved paths; must reduce training sequential depth to $O(\log N)$ or at least $O(N)$; must preserve polar code structure; must scale to $N=256/512/1024$). [filecite\turn0file1](#)

Executing the searches implied by those documents yields several high-value anchor results and adjacent bodies of work:

- For the *analytical foundations and the exact bottleneck*, the most relevant publicly available “computational graph / monotone chain” reference found is a 2025 arXiv preprint on general monotone-chain polar codes. It explicitly analyses how chain structure affects decoding time complexity (from $O(N \log N)$ up to $O(N^2)$) and introduces a computational-graph framing. [1](#)
- For *MAC polar-code design and decoding on Gaussian MACs* (including “joint successive cancellation” vs “joint iterative” approaches and frozen-bit design issues), an arXiv preprint from 2019 directly targets the GMAC with polar codes and compares decoding techniques. [2](#)
- For *neural polar decoders and training methodology on unknown / memory channels* (the closest directly relevant “neural SC-style” lineage), there are IEEE-indexed works by Ziv Aharoni [3](#) and collaborators (conference + journal). [4](#)
- For *parallel training alternatives that preserve code structure*, two externally validated directions have strong precedent: (i) learned/weighted belief-propagation style decoders (including the seminal “learned BP” line by Eliya Nachmani, and later learned BP for polar codes), and (ii) transformer-based decoders for polar codes (e.g., a 2025 arXiv transformer polar decoder). [5](#)
- For the *Walsh-Hadamard / character-domain decomposition* direction, there is a primary peer-reviewed source showing the dyadic-convolution $\leftrightarrow$ pointwise-multiplication theorem under Walsh-Hadamard transforms, which is conceptually aligned with the “diagonalise convolution on $Z_2 \times Z_2$ ” approach but does not by itself resolve log-domain/decision-order issues. [6](#)

Overall, the web evidence supports a key conclusion: the “parallelisation bottleneck” described in the files is not merely an engineering inconvenience; it aligns with known complexity sensitivity to monotone-chain structure (especially alternating/interleaved chains). [1](#) The most credible paths to practical parallel training that *still respect polar structure* are (a) moving to parallel message-passing/unrolled iterative decoding (BP-like, potentially learned), and (b) “domain changes” (e.g., WHT/character

domain) combined with *leaving strict log-domain tensor algebra*, or accepting principled approximations to log-sum-exp arithmetic.

Extracted instructions from the provided files

The table below consolidates *explicit* and *implied* research/search instructions, keywords, priorities, constraints, and any stated time windows. `filecite{turn0file0}` `filecite{turn0file1}`

File	Explicit research/ search tasks	Search keywords, targets, and entities (explicit or implied)	Priorities	Constraints and filters (explicit)
full_project_report.pdf	Conduct and report a literature survey; identify gaps; position the work vs. prior art; define future directions including “closing $N \geq 256$ gap” and “parallel training for MAC”. <code>filecite{turn0file0}</code>	Neural SC decoding; polar codes; two-user MAC; monotone chain paths (Class B emphasis); (CalcLeft/ CalcRight/ CalcParent); circular convolution over $Z_2 \times Z_2$; z_{encoder} bottleneck; scaling to larger N; channels: BEMAC, GMAC, ABNMAC, ISI-MAC; key related-work clusters: neural polar decoders; analytical MAC polar; DNN multi- user detection; end-to-end learned MAC codes. <code>filecite{turn0file0}</code>	“Closing the $N \geq 256$ gap” is labelled <i>highest priority</i> and “Parallel training for MAC” is labelled <i>highest impact</i> . <code>filecite{turn0file0}</code>	Literature search window is explicitly stated as 2017– 2026 (for “60+ papers”); emphasis on practical symmetric-rate paths (Class B); focus on continuous channels for scaling failure. <code>filecite{turn0file0}</code>

File	Explicit research/ search tasks	Search keywords, targets, and entities (explicit or implied)	Priorities	Constraints and filters (explicit)
gemini_research_prompt.pdf	Understand why MAC training is sequential; understand why fast_ce works in single-user but fails for MAC; search deeply for decompositions/ alternative paradigms; look across fields (group theory, tensor decompositions, parallel algorithms, tropical algebra); propose implementable approaches; provide ranked approach list + feasibility analysis; provide pseudocode for top approach. □filecite□turn0file1□	“parallelize training”; “fast_ce”; “Class B / interleaved”; “circular convolution over $Z_2 \times Z_2$ (Klein four-group)”; “Walsh–Hadamard transform”; “log domain / log-probability”; “logsumexp”; “tropical semiring / max-log”; “factor graph / belief propagation”; “transformers / attention”; “tensor decompositions”; “group-theoretic circuits” (Z_2^k); “curriculum of parallelism”; “two-phase / iterative refinement.” □filecite□turn0file1□	Primary objective: reduce training sequential depth from $O(N \log N)$ to $O(\log N)$ (or accept $O(N)$). Must work for Class B. □filecite□turn0file1□	Must: (i) match/ approach analytical SC at convergence; (ii) $O(\log N)$ or $O(N)$ sequential depth during training; (iii) work for Class B; (iv) be compatible with teacher forcing; (v) scale to $N=256/512/1024$. □filecite□turn0file1□

Consolidated extraction of constraints, timeframes, domains, and data types

The following items are either stated directly or strongly implied across the two files:

- Timeframes explicitly stated: the literature search in the project report claims a review spanning 2017–2026. □filecite□turn0file0□
- Timeframes implied for *technical targets*: up-scaling to $N=256/512/1024$ is repeatedly a key benchmark threshold. □filecite□turn0file0□ □filecite□turn0file1□
- Domain constraints: focus on polar codes for *two-user MAC*; focus on *Class B / interleaved* monotone chain paths that achieve symmetric rate points; prefer solutions that preserve polar structure. □filecite□turn0file0□ □filecite□turn0file1□
- Data types to search: peer-reviewed papers, arXiv preprints, standards documents (for context), algorithmic/theoretical papers on group convolution and log-semirings, and (where available) code repositories for reproducibility. □filecite□turn0file0□ □filecite□turn0file1□
- Geographic constraints: none specified. (Unspecified in both files.) □filecite□turn0file0□ □filecite□turn0file1□

- Language preference: not stated in the files, but the user’s instruction is to prefer English sources.

Search strategy and query log

Prioritised source types

The searches were executed by prioritising primary and official sources wherever accessible (open preprints, official conference pages, standards bodies, and direct journal hosting). When the canonical publisher page was blocked or inaccessible, bibliographic mirrors (university repositories, DOAJ, DBLP, or institutional research portals) were used as secondary evidence.

Source (prioritised)	Why it was prioritised	Typical evidence retrieved	Notes on applicability to the files’ tasks
arXiv ⁷	Primary source for open-access original papers, especially recent technical work	Full abstracts, submission dates, PDF access, author lists	High value for “Ren et al. 2025” and GMAC polar code work; supports quick verification of existence and scope ⁸
ETSI ⁹ / 3GPP ¹⁰	Official standards texts (context for “polar codes in 5G”)	Full PDFs describing polar coding procedures	Useful context only; not central to the MAC-parallelisation problem ¹¹
ICML ¹²	Official ML venue pages; direct access to posters/PDFs	Poster pages and paper PDFs for neural decoders (CRISP/DeepPolar)	Provides verified references for sequential/curriculum neural polar decoders ¹³
NeurIPS	Official venue pages, especially for workshop outputs	Workshop poster pages	Helps validate “NPD for 5G systems” direction ¹⁴
DOAJ ¹⁵ and journal-hosted PDFs	Verifiable open-access copies of older, peer-reviewed work	Direct PDFs of convolution/WHT results	Supports Direction 1 (WHT diagonalisation) with a stable published PDF ¹⁶
University repositories/ research portals	Often provide stable metadata + DOI when publisher pages are blocked	Publication metadata, abstracts, sometimes PDFs	Used for polar/MAC references and learned-BP lineage when needed ¹⁷

Workflow and timeline diagrams

flowchart TD

A[Extract tasks & constraints from both PDFs] --> B[Derive structured query sets]

B --> C[Run source-first searches: arXiv, official venues, standards bodies]

C --> D[Backfill with bibliographic mirrors if publisher pages blocked]

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D --> E[Cluster findings by the prompt's research directions]
E --> F[Synthesise evidence into tables + gap analysis + next steps]
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```
gantt
  title Research execution timeline
  dateFormat YYYY-MM-DD
  axisFormat %d %b
  section Extraction
  Parse & normalise tasks from files :done, 2026-04-04, 1d
  section Search
  Primary source sweep (arXiv/venues/standards):done, 2026-04-04, 1d
  Secondary source sweep (mirrors/repositories):done, 2026-04-04, 1d
  section Synthesis
  Evidence clustering + reporting :done, 2026-04-04, 1d
```

Search queries actually used

The executed search queries were grouped to map directly onto (a) the explicit references listed in the files, and (b) each “Direction to Explore” in the research brief. The log below is representative and covers all major query families:

Query family	Example queries (verbatim category)	Purpose	Representative sources found
Foundational polar/MAC references	“polar codes for the two-user multiple-access channel Sasoglu Telatar Yeh 2013”, “Successive cancellation decoding ... two-user binary-input MAC Onay 2013”	Verify and retrieve primary references for polar MAC + SC decoding	Two-user MAC polar codes paper metadata ¹⁸ ; Onay 2013 repository PDF listing ¹⁹
“Ren et al. 2025” computational-graph decoder	“Successive Cancellation Decoding For General Monotone Chain Polar Codes arXiv 2509.03128”	Identify the computational-graph / monotone-chain paper and its complexity claims	arXiv entry + abstract with complexity characterisation ¹
GMAC polar design/decoding	“Design and Decoding of Polar Codes for the Gaussian Multiple Access Channel arXiv 1901.07297”	Find analytical MAC decoding variants and frozen-bit design methods	arXiv entry describing “joint SC” and “joint iterative” algorithms ²
Neural polar decoder lineage	“Data-driven Neural Polar Decoders for Unknown Channels with and without Memory 10.1109/TIT. 2024.3476681”	Anchor the NPD/ NSC family in primary sources	Bibliographic page with IEEE doc id + metadata ²⁰

Query family	Example queries (verbatim category)	Purpose	Representative sources found
Curriculum/ alternative neural decoders	"CRISP curriculum based sequential neural decoders polar code family ICML 2023", "DeepPolar Inventing Nonlinear Large-Kernel Polar Codes ICML 2024"	Map the broader landscape of neural polar decoders to what the project report claims it surveyed	ICML official poster pages for CRISP/DeepPolar ¹³
Learned BP / message passing	"Learning to decode linear codes using deep learning Nachmani arXiv 1607.04793", "Deep Learning Methods for Improved Decoding of Linear Codes 10.1109/JSTSP.2017.2788405"	Validate the "weighted BP with learned parameters" direction	arXiv for learned BP ²¹ ; journal metadata (2018) ²²
Transformer decoders for polar codes	"transformer polar decoder arXiv", "latent-attention transformer polar decoding 2507.14951"	Validate whether transformer polar decoders exist and what claims they make	arXiv transformer polar decoder entry ²³
WHT/character-domain convolution theorem	"Walsh-Hadamard Transformation of a Convolution Radioengineering PDF"	Find a peer-reviewed source formalising the WHT diagonalisation of dyadic convolution	Radioengineering hosted PDF + issue page ⁶
Log-domain approximations / max-log	"max* operator Log-MAP approximation", "reduced-complexity belief propagation decoding for polar codes approximation"	Support the "tropical / max-log" direction with established decoding approximations	Log-MAP approximation paper ²⁴ ; approximate BP polar decoding ²⁵

Results and evidence

The findings below are organised to mirror the research brief's central directions and the project report's stated priorities. Each entry includes evidence and why it matters.

Evidence table of findings

Finding (what was found)	What it provides (evidence)	Relevance to the files' objectives
Two-user MAC polar codes are formally established as an extension of polar coding to MACs	Publication metadata confirms a peer-reviewed IEEE Transactions on Information Theory article on polar codes for the two-user MAC, including capacity-region statements and $O(n \log n)$ -style scaling claims in its abstract/metadata	Foundational for both files' premise that polar codes are the structured base code for two-user MAC research ¹⁸
SC decoding for two-user binary-input MAC using monotone chain rules exists in the ISIT 2013 literature	A university repository entry provides title/abstract describing SC decoding for two-user binary-input MAC with monotone chain expansions and rate-region goals	Supports the "Analytical SC decoder exists but is sequential" piece of the brief and project report ¹⁹
Monotone-chain / computational-graph SC decoding generalisations exist and explicitly attribute complexity variation to chain structure	The arXiv 2025 paper states SC decoding time can range between $O(N \log N)$ and $O(N^2)$ depending on decoding chain structure; also motivates a computational graph framework	This directly backs the brief's claim that alternating/interleaved paths create cross-level dependencies, and suggests the bottleneck is structurally real rather than an implementation detail ¹
GMAC-specific polar-code design and decoding work discusses joint SC and joint iterative techniques and frozen-bit design	The GMAC polar arXiv paper abstract details non-orthogonal multiple access, compares "joint successive cancellation" and "joint iterative" algorithms, and proposes a design algorithm for frozen bits	Aligns with the project report's GMAC focus (continuous channels) and offers alternative decoding paradigms that are inherently more parallel (joint iterative) ²
Data-driven neural polar decoding for unknown channels (and with memory) is an active, IEEE-indexed line of work	Evidence includes an IEEE Transactions on Information Theory paper record (with IEEE doc link) describing "data-driven neural polar decoders" for channels with/without memory	Provides validated precedent for replacing analytical SC operations with learned modules, and may offer transferable training ideas (e.g., sufficient statistic learning) but does not solve MAC Class B parallelisation by itself ²⁶
Curriculum-based sequential neural polar decoders exist (single-user), showing curriculum learning as an established training tool	ICML official CRISP poster page describes a sequential neural decoder trained via a curriculum, outperforming SC on short polar codes	Supports the project report's use of curriculum ideas; also evidences Direction 10 ("curriculum of parallelism") as plausible, though CRISP remains sequential (not the parallel breakthrough sought) ²⁷

Finding (what was found)	What it provides (evidence)	Relevance to the files' objectives
Learned/weighted belief propagation decoding is a mature paradigm for neuralising message passing	arXiv preprint describes learning weights on Tanner-graph edges for improved BP decoding; a later peer-reviewed journal entry describes improving BP/min-sum and parameter tying (RNN-like)	Directly supports Direction 3 (factor graph / message passing) and provides a strong template for “parallel-by-iteration” training: iterations give controllable sequential depth while keeping parallel node updates ²⁸
Learned BP ensembles specifically for polar codes exist	A conference publication record describes CRC-aided learned ensembles of BP polar decoders and claims measurable SNR/BER gains	Strengthens the “BP-style as a practical parallel alternative within polar structure” argument: it has been tried on polar factor graphs (even if not in MAC) ²⁹
Transformer-based polar decoding exists (short-code regime)	A 2025 arXiv paper describes a latent-attention transformer for polar decoding and reports near-ML performance claims for short codes	Validates Direction 4 (“transformer-based parallel decoder”) as a real research direction in polar decoding, though scaling to long N and generalisation to MAC remain open risks ²³
The Walsh–Hadamard transform diagonalises dyadic convolution (convolution theorem) in a peer-reviewed, open-access PDF	The Radioengineering paper and its official hosting show that (dyadic) convolution becomes coordinate-wise multiplication in the Walsh–Hadamard domain	Provides hard evidence for Direction 1’s mathematical claim: convolution over a binary group can become separable after transform. However, it does not automatically resolve the brief’s noted log-domain + CalcRight + path-order problems ⁶
There is established literature on approximating log-domain “sum” operations (max* / max-log type ideas) in decoding-related algorithms	An open-access Log-MAP approximation paper formalises cheaper approximations to the log-sum-exp-like max* operations; approximate BP polar decoding papers exist	Supports Direction 2: principled approximations exist that reduce expensive log-domain operations. It still requires careful adaptation to the $Z_2 \times Z_2$ tensor setting and to training stability/accuracy requirements ³⁰
Recent work explicitly studies NPD-type decoders in more realistic comms systems (5G-oriented)	A NeurIPS workshop page describes adaptation and analysis of neural polar decoders for end-to-end systems, including rate matching and robustness concepts	Adds evidence that NPD research is moving toward systems-grade constraints, which may affect how a parallel training decomposition should be evaluated (hardware and rate-matching constraints) ¹⁴

Finding (what was found)	What it provides (evidence)	Relevance to the files' objectives
Standards evidence for the "polar codes in 5G NR" claim is publicly accessible	ETSI TS 138 212 PDFs contain polar coding clauses and are official spec artefacts	Useful to ground "why this matters" context (polar decoders are not purely academic), though it does not directly constrain the MAC parallel-training task ¹¹

Visual grounding for key sources and concepts

Analytical synthesis mapped to the research-brief directions

The research brief's "Directions to Explore" (WHT, tropical/max-log, BP, transformers, etc.) can be tied to concrete, externally evidenced work as follows:

- Direction on WHT/character-domain diagonalisation: a published, open-access theorem exists for dyadic convolution and Walsh–Hadamard transforms, providing a mathematically correct backbone for "convolution $\rightarrow$ element-wise multiplication" claims. ⁶ The missing piece remains exactly what the brief states: the MAC decoder's pipeline includes non-convolution steps and (in typical implementations) log-domain calculations. `fileciteturn0file1`
- Direction on tropical/max-log approximations: decoding literature routinely approximates expensive "log-sum-exp-like" operations with cheaper max*/max-log style operators (sometimes with correction terms), and approximate BP polar decoders have been published. ³⁰ This suggests a viable search path for "replace logsumexp with an approximation that may be separable enough to parallelise," but the approximation error vs. "match analytical SC at convergence" constraint is not yet bounded in the MAC $Z_2 \times Z_2$ tensor setting (unresolved gap). `fileciteturn0file1`
- Direction on BP/message passing: BP decoders for polar codes are widely reviewed and explicitly noted for parallelism, and there is at least one learned-BP polar-decoder ensemble paper. ³¹ Combined with the learned-BP lineage for general linear codes, this is the strongest evidence-backed route to *parallel training without destroying code structure*. ²⁸
- Direction on transformer decoders: transformer-based polar decoders are real and recent, at least for short-code regimes. ²³ This supports "fully parallel training/inference" in principle, but the brief explicitly disallows vague "just use transformers" answers; the key remaining question is whether teacher-forced training meaningfully transfers to the MAC interleaved SC dependency structure without collapsing generalisation. `fileciteturn0file1`
- Structural validation of the "interleaved path makes it hard" claim: the arXiv monotone-chain paper explicitly links chain structure to complexity blow-ups, consistent with the brief's "not level-aligned" dependency story. ¹

Gaps and recommended next steps

Gaps in constraints and specification

Several constraints that are critical for a rigorous "parallel training breakthrough" are not specified in the files and should be treated as genuinely unspecified:

- Quantitative acceptance criteria: no explicit threshold is given for what "matches analytical SC" means (BLER delta at which SNR, which N, which number of trials). (Unspecified.) `fileciteturn0file1`

- Target channel/model priority for validating a parallel method: the materials discuss both discrete (BEMAC-like) and continuous (GMAC) channels, but the brief does not declare which is the *required* benchmark for the parallelisation proposal. (Unspecified.)
- Compute/hardware constraints for the proposed method: batch size, GPU/CPU assumptions, memory limits, and training-time budget are not standardised. (Unspecified.)
- Scope of “preserve polar structure”: the brief disallows “abandoning polar code structure entirely,” but it does not specify whether factor-graph BP decoding (which uses the generator/factor graph rather than SC tree-walk) counts as “preserving structure” (it plausibly does). (Ambiguous/unspecified.)
- Defined search venues/databases: neither file mandates a specific database (e.g., IEEE Xplore, ACM DL) or a required completeness guarantee for literature coverage. (Unspecified.)

Recommended next steps grounded in the evidence

The following next steps follow directly from the extracted tasks and the strongest evidence found, and are framed to satisfy the brief’s constraints (Class B, teacher forcing compatibility, preserve structure) as closely as possible.

First, treat “parallel training” as two separate sub-problems:

- replacing the *sequential SC tree-walk* with a *parallel update schedule* (e.g., BP/unrolled iterations), and
- retaining enough of the polar/MAC structure that convergence can approach analytical SC performance.

This decomposition is supported by the presence of mature message-passing and learned-BP literature (general codes + polar-specific). ³²

Second, use the monotone-chain computational-graph paper as a constraints oracle: explicitly map which dependency patterns force $O(N^2)$ -like behaviour under alternating chains, and design any “parallel schedule” to avoid those dependency-reintroduction patterns. ¹

Third, prioritise two research routes for implementation/testing (because they already have strong external precedent):

- Learned BP / unrolled-message-passing for MAC polar codes.
This is the most evidence-backed “parallel-by-iteration” candidate: (i) BP updates are inherently parallel within each iteration; (ii) learned-BP shows parameters can be trained and tied across iterations; (iii) polar-code BP variants and learned ensembles exist, de-risking the “polar structure compatibility” objection. ³³
- WHT/character-domain re-parameterisation *outside* strict log-domain tensor algebra.
The WHT dyadic convolution theorem is real and peer-reviewed; the key engineering research task becomes: can the MAC SC operations be reformulated to keep computations in a domain where diagonalisation helps without reintroducing non-separable logsumexp bottlenecks? ⁶
The brief itself notes the log-domain obstruction; this route is promising only if one changes the representation or uses stable approximations with bounded loss.

Fourth, treat transformer decoders as a controlled baseline rather than *the* answer. The existence of transformer polar decoders supports feasibility but does not solve the “structured dependency”

argument by itself. ³⁴ A disciplined next step is: define a teacher-forced, mask-based transformer objective that emulates the Class B dependency graph, then evaluate whether it generalises beyond the training SNR/channel class.

Finally, close the loop with the project report's highest-priority and highest-impact tasks by formalising success metrics that directly test them:

- "Closing $N \geq 256$ gap" should be evaluated on continuous-output channels (GMAC-like) and on the N threshold explicitly called out in the report and brief. [\[filecite turn0file0\]](#) [\[filecite turn0file1\]](#)
- "Parallel training for MAC" should be evaluated with a wall-clock + sequential-depth metric (e.g., critical path length per forward/backprop) in addition to decoding performance, because the goal is explicitly training-time reduction ("days to hours"). [\[filecite turn0file0\]](#)

¹ ⁸ ⁹ <https://arxiv.org/abs/2509.03128>
<https://arxiv.org/abs/2509.03128>

² <https://arxiv.org/abs/1901.07297>
<https://arxiv.org/abs/1901.07297>

³ ²³ ³⁴ <https://arxiv.org/abs/2507.14951>
<https://arxiv.org/abs/2507.14951>

⁴ <https://cris.iucc.ac.il/en/publications/data-driven-polar-codes-for-unknown-channels-with-and-without-mem/>
<https://cris.iucc.ac.il/en/publications/data-driven-polar-codes-for-unknown-channels-with-and-without-mem/>

⁵ ¹² ¹⁵ ²¹ ²⁸ ³² <https://arxiv.org/abs/1607.04793>
<https://arxiv.org/abs/1607.04793>

⁶ <https://dspace.vut.cz/items/0d43cdee-898a-48d8-a816-67bfa99fac08>
<https://dspace.vut.cz/items/0d43cdee-898a-48d8-a816-67bfa99fac08>

⁷ ¹³ ²⁷ <https://icml.cc/virtual/2023/poster/24889>
<https://icml.cc/virtual/2023/poster/24889>

¹⁰ ³¹ <https://www.mdpi.com/2073-8994/14/12/2633>
<https://www.mdpi.com/2073-8994/14/12/2633>

¹¹ https://www.etsi.org/deliver/etsi_TS/138200_138299/138212/18.02.00_60/ts_138212v180200p.pdf
https://www.etsi.org/deliver/etsi_TS/138200_138299/138212/18.02.00_60/ts_138212v180200p.pdf

¹⁴ <https://neurips.cc/virtual/2025/123181>
<https://neurips.cc/virtual/2025/123181>

¹⁶ <https://doaj.org/article/e758bdc6d06645d693296093990f6fd4>
<https://doaj.org/article/e758bdc6d06645d693296093990f6fd4>

¹⁷ ¹⁹ <https://repository.bilkent.edu.tr/items/ee022b06-cbc1-432e-a02d-3e3ed3954463>
<https://repository.bilkent.edu.tr/items/ee022b06-cbc1-432e-a02d-3e3ed3954463>

¹⁸ <https://infoscience.epfl.ch/entities/publication/747e89a8-5c7a-4a54-a817-2c9240dcbb4a>
<https://infoscience.epfl.ch/entities/publication/747e89a8-5c7a-4a54-a817-2c9240dcbb4a>

²⁰ ²⁶ <https://colab.ws/articles/10.1109/tit.2024.3476681>
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22 33 <https://cris.tau.ac.il/en/publications/deep-learning-methods-for-improved-decoding-of-linear-codes>

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24 30 <https://link.springer.com/article/10.1186/1687-1499-2013-238>

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25 https://www.jstage.jst.go.jp/article/transfun/E100.A/9/E100.A_2052/_article/-char/en

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29 <https://cris.technion.ac.il/en/publications/crc-aided-learned-ensembles-of-belief-propagation-polar-decoders/>

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